Girijananda Chowdhury University, Guwahati 781017, Assam

School of Engineering and Technology, Civil Engineering Department

Formative Assessment – I (5th Semester)

CE-BCE23504T – Hydrology and Water Resources Engineering

10th October, 2026
Duration: 1 hour

Total Marks: 20

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1. Diagrammatically describe the hydrologic cycle. **(8)**
2. Define Ø-index. **(2)**
3. The average rainfall over 55 hectares of watershed for a particular storm are as follows- **(5)**

Time (hr)	0	1	2	3	4	5	6	7
Rainfall (cm)	0	0.75	1.25	3.40	2.95	1.40	0.80	0

The volume of run-off from the storm was determined a 3.20 ha-m. Establish the Ø - index.

4. The total observed runoff volume during a 6 hours storm with a uniform intensity of 1.5 cm/hour is 21.6×10^6 m^3. If the area of the basin is 300 km^2, find the average infiltration rate for the basin and depth of runoff. **(5)**